

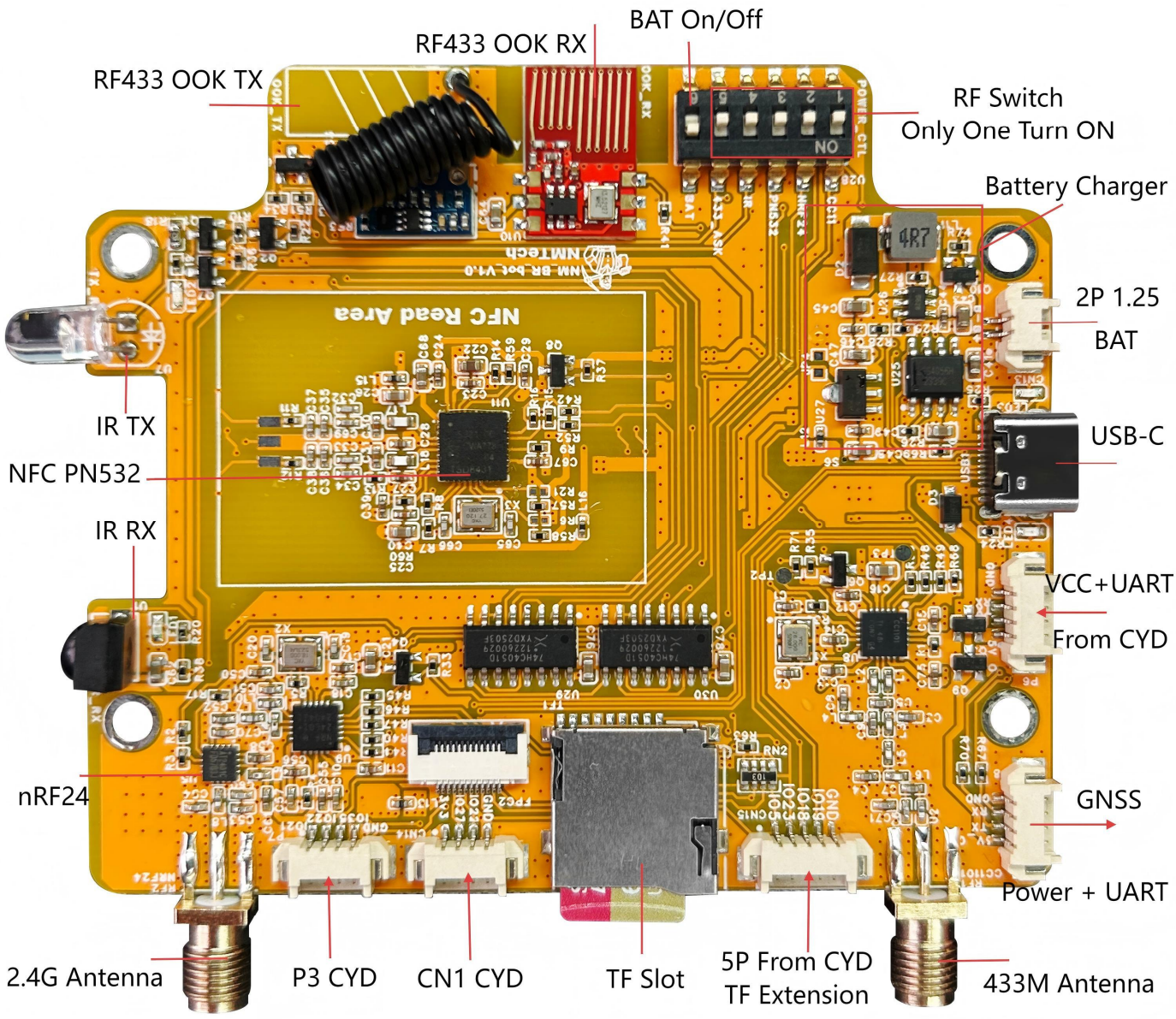
NM-RF-HAT
User manual

1 Specification

1.1 General Specification

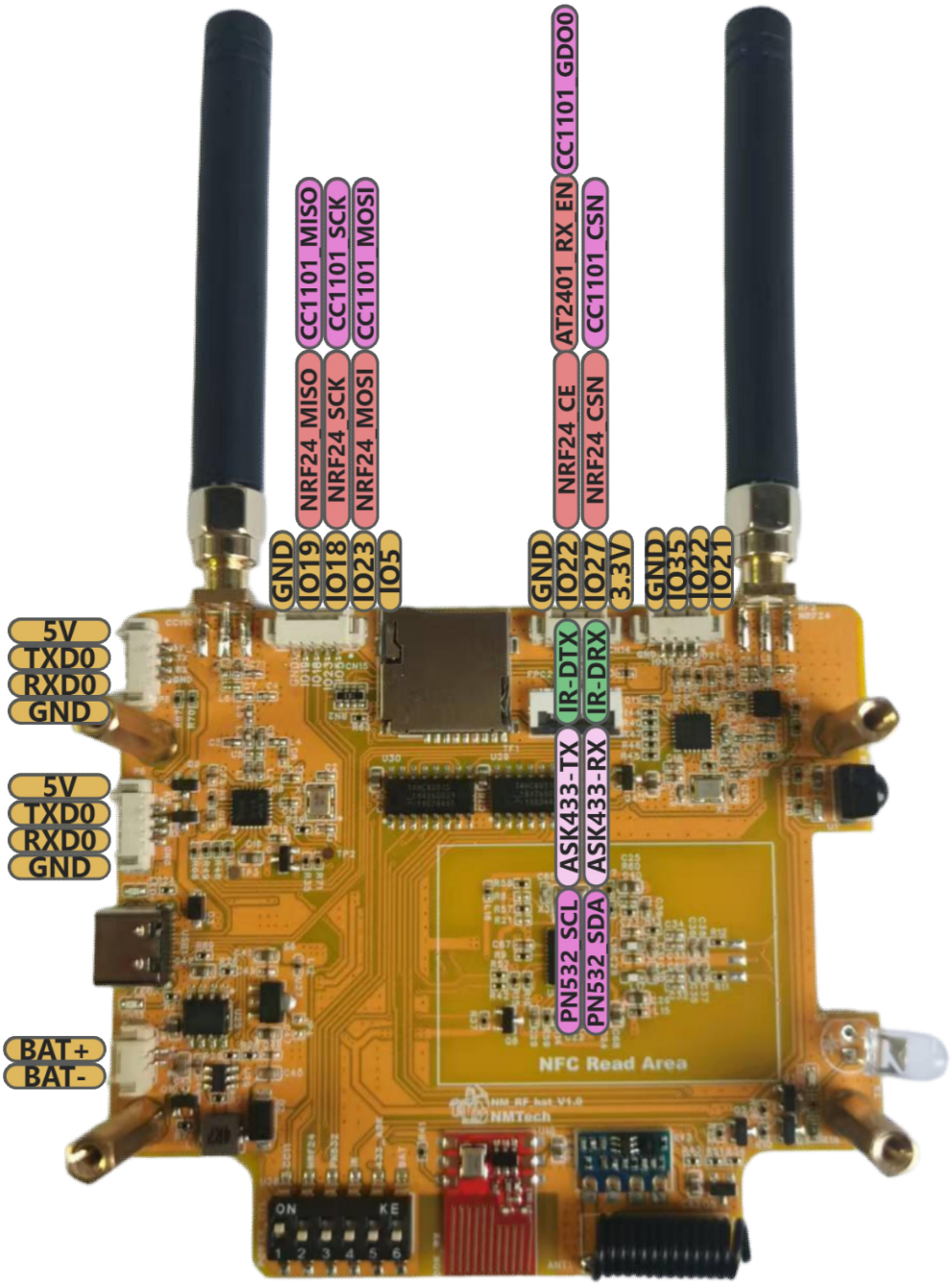
Title	Info	
Module	433MHz FSK RF transceiver	CC1101
	2.4GHz FSK RF transceiver	NRF24L01
	13.56M NFC controller	PN5321
	RF433-ASK/OOK transceiver	Discrete component circuits
	IR control transceiver	Discrete component circuits
Supply mode	USB 5V DC / Battery 3.7V	
Supply Interface	Type-C USB(5V DC) /MX1.25 connector （Battery ）	
Function switch	DIP switch	
Battery voltage range	2.5V~4.2V	
Battery charge current	500mA	
CC1101 433M RF connector	External thread internal pin SMA	
NRF24 2.4G RF connector	External thread internal hole SMA	
Power Consumption	1~1.9W(Assembled with CYD)	
Operating Temperature	-20 ~ 70 ℃	

1.2 Function diagram and description



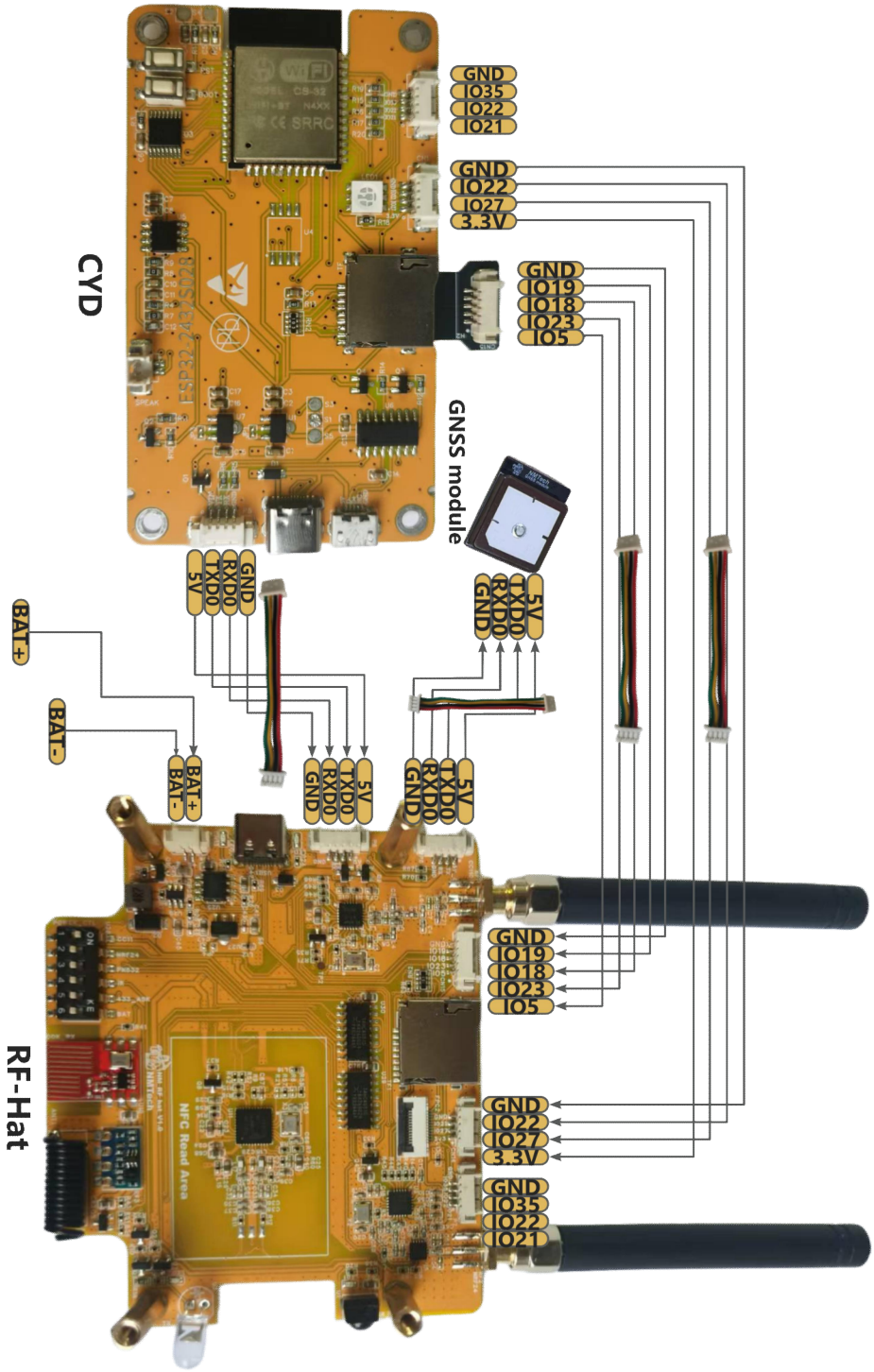
Module	Frequency /Type	Function	Key Benefit
CC1101	433MHz	Remote Control, Sensor Capture	Comprehensive Sub-GHz analysis and replay.
NRF24	2.4GHz	Wireless Protocol Analysis	Targeting common 2.4G peripherals (mice, keyboards).
PN532	NFC/RFID	Read, Write, Emulation	High-frequency RFID access and cloning capabilities.
IR Control	IR TX/RX	Infrared Signal Capture/Replay	Universal remote control and analysis.
RF433	OOK ASK 433MHz	Basic RF Transmission	Simple 433MHz OOK signal handling.
Power management	USB-C, BAT Circuit	Charging, On/Off Switch	Stable, portable power for field use.
GPS Expansion	GPS Header	GPS/Location Tracking	Used for Wardriving and GPS tracking.

1.3 Pin map

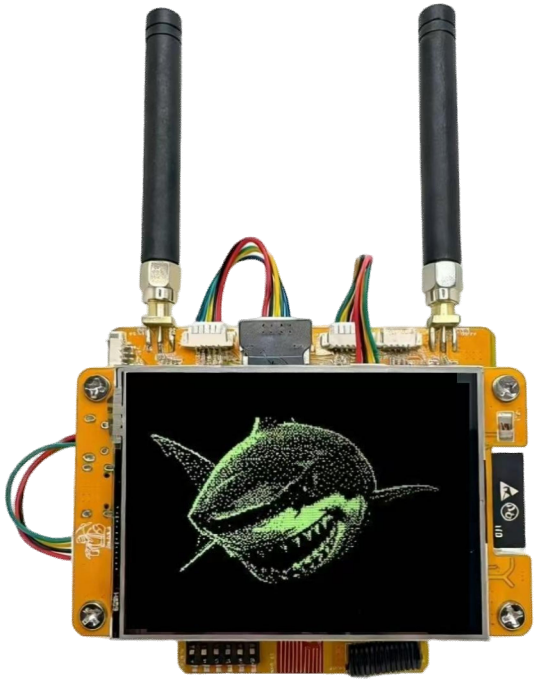

CC1101_GDO0
AT2401_RX_EN
CC1101_CSN
CC1101_MISO
CC1101_SCK
CC1101_MOSI
NRF24_CE
NRF24_CSN
NRF24_MISO
NRF24_SCK
NRF24_MOSI
GND
IO19
IO18
IO23
IO5
GND
IO22
IO27
3.3V
GND
IO35
IO22
IO21
5V
TXD0
RXD0
GND
IR-DTX
IR-DRX
5V
TXD0
RXD0
GND
ASK433-TX
ASK433-RX
PN532_SCL
PN532_SDA
BAT+
BAT-
NFC Read Area
NMTech

2 Assemble with CYD

2.1 Connection diagram



2.2 Assembled unit





3 Get Start and configuration

3.1 Module switch

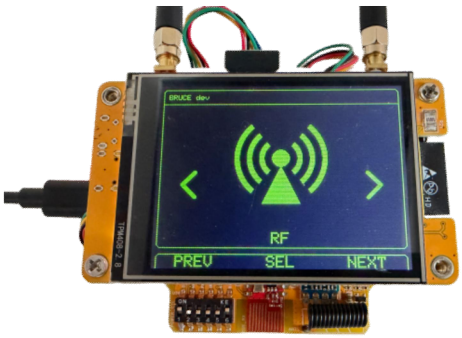
The NM-RF-HAT uses a 6-position DIP switch for function switching. The sixth position is the battery power switch, while position 1 to 5 are function switches. Only one function switch can be set to the ON, and only one function can be active at any given time.



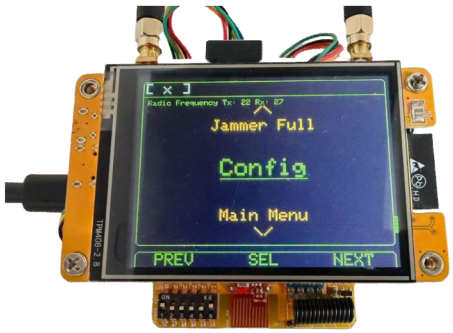
3.2 CC1101(433M)

When you start using the CC1101 function, you will follow the steps below.

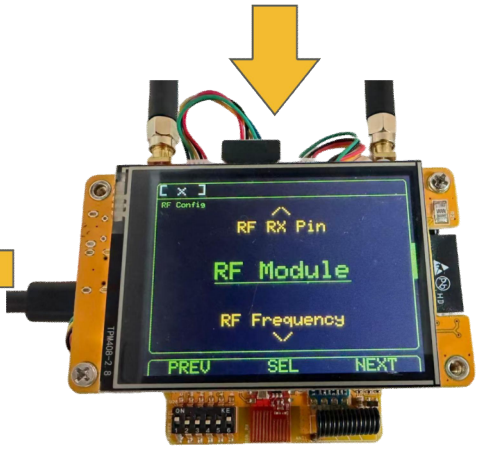
- Set the DIP switch to position 1 and select the RF option.
- Choose the Config option.
- Select the RF Module option.
- Pick the CC1101 option.
- Then return to RF menu to use other CC1101 functions.



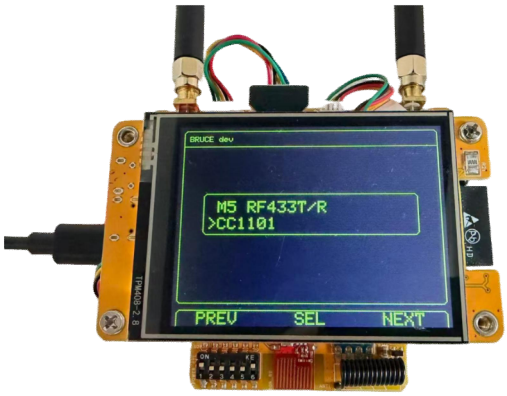
1:Set the DIP switch to position 1 and click the RF option.



2:Select the config option



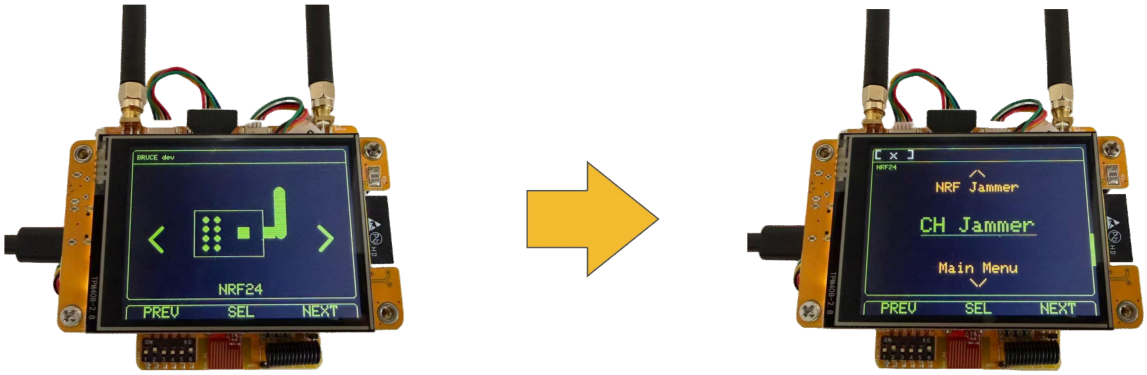
3:Select the RF Module option



4:Select the CC1101 option

3.3 NRF24

- Set the DIP switch to position 2 and select the NRF24 option.
- Select the NRF functions (such as Spectrum, NRF Jammer, or CH Jammer) as needed.



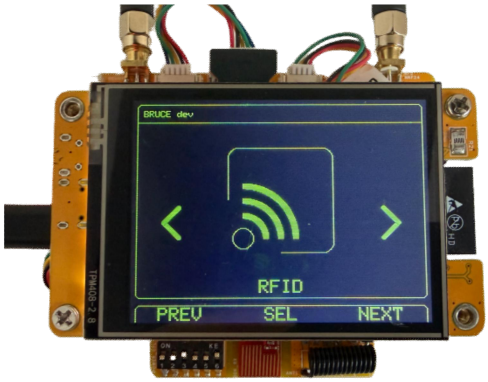
1:Set the DIP switch to position 2 and click the NRF24 option.

2:Select the NRF Functions

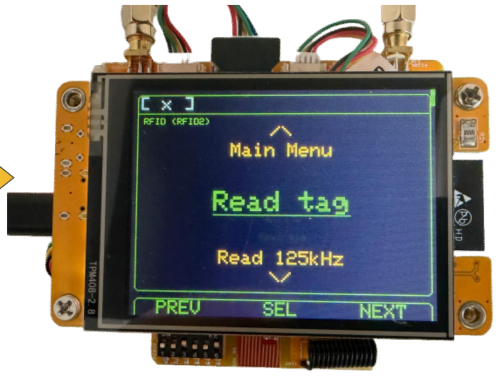
3.4 PN532 NFC

When you start using the RFID/NFC function, you will follow the steps below.

- Set the DIP switch to position 3 and select the RFID option (Config -> RFID module -> choose PN532 on I2C).
- Select the RFID functions (such as Read tag) as needed.



1:Set the DIP switch to position 3 and click the RFID option.

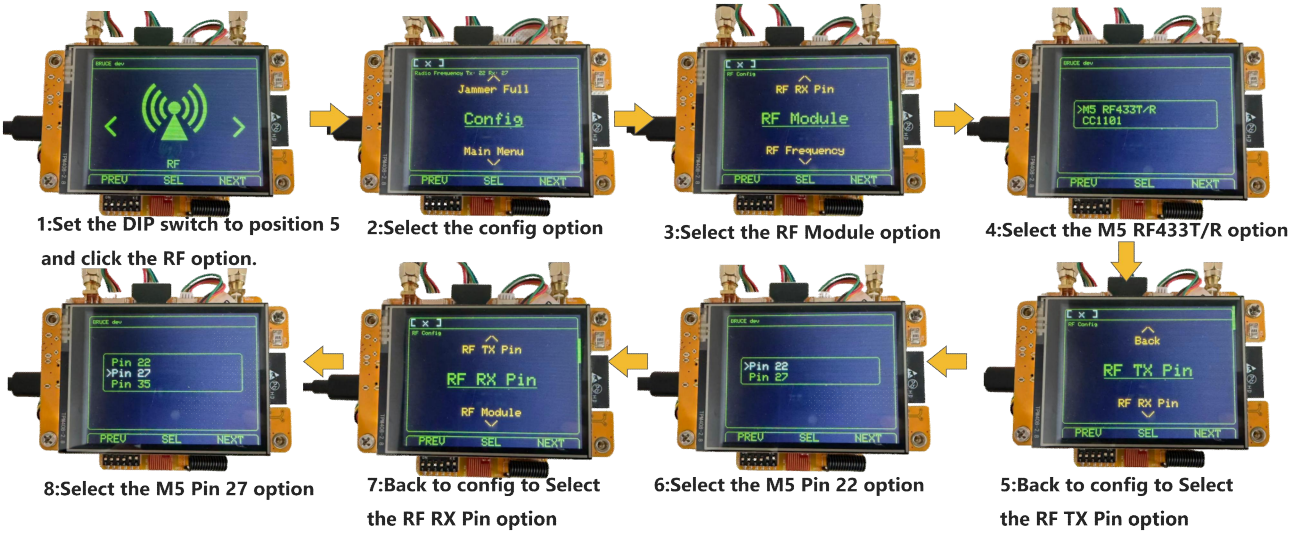


2:Select the RFID Functions

3.5 RF 433M -OOK/ASK

When you start using the RF 433M -OOK/ASK function, you will follow the steps below.

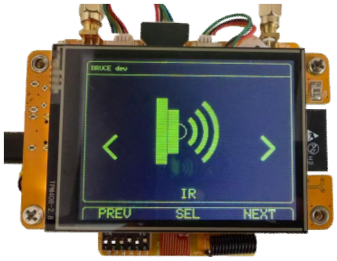
- Set the DIP switch to position 5 and click the RF option.
- Select the Config option.
- Select the RF Module option.
- Select the M5 RF433T/R option.
- Return to the config menu and select the RF TX Pin option.
- Select the M5 Pin 22 option.
- Return to the config menu and select the RF RX Pin option.
- Select the M5 Pin 27 option.
- Then return to RF menu to use other RF433 functions.



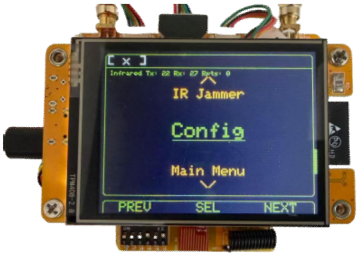
3.6 IR controller

When you start using the IR controller function, you will follow the steps below.

- Set the DIP switch to position 4 and select the IR option.
- Select the Config option.
- Select the IR TX Pin option.
- Select the Pin 22 option.
- Return to the config menu and select the Ir RX Pin option.
- Select the Pin 27 option
- Then return to IR menu to use other IR functions.



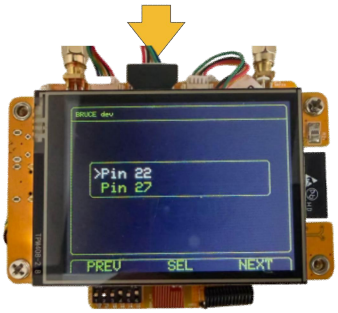
1:Set the DIP switch to position 4 and click the IR option.



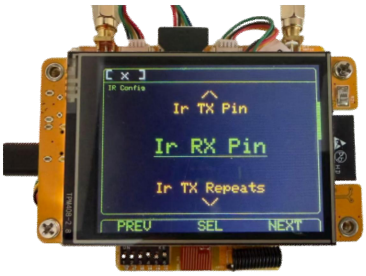
2:Select the Config option.



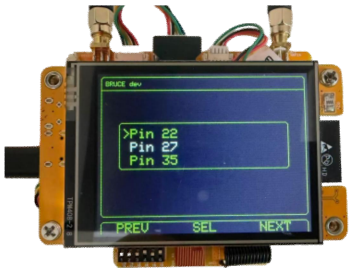
3:Select the Ir TX Pin option.



4:Select the Pin 22 option.



5:Select the Ir RX Pin option.

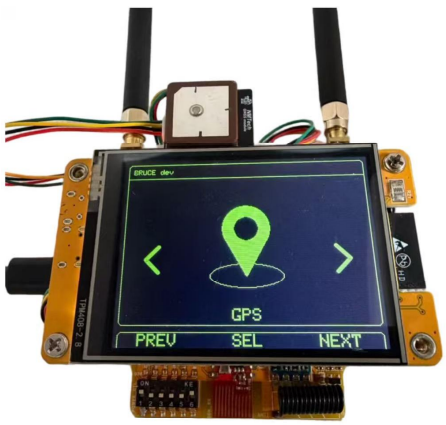


6:Select the Pin 27 option.

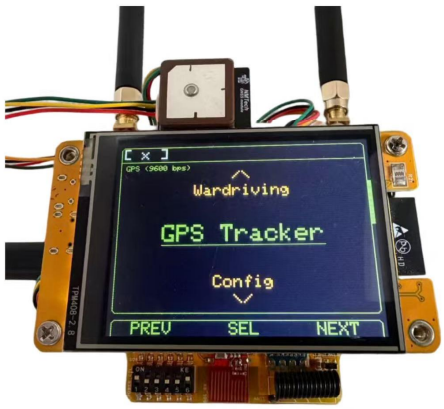
3.7 GPS

When using the GPS function, you need to connect the GPS module first, and then follow the steps below to operate.

- Select the GPS option (the official firmware changed default pins to RX=22 TX=27, for nm-rf-hat change to RX=1 TX=3)
- Select the GPS function such as GPS Tracker/Wardriving



1:Select the GPS option



2:Select the GNSS function such as GPS Tracker

4.Contact NM-Tech

Bruce Wiki: https://wiki.bruce.computer/

Home page: https://www.nmminer.com

Telegram: https://t.me/NMMiner

Reddit: https://www.reddit.com/r/NMTech_Team/

Email: nmminer1024@gmail.com

AliExpress (NMTech Global Store): https://www.aliexpress.com/store/1104268817

YouTube Channel